



UWLight-YOLO: An adaptive lightweight attention network for underwater sonar image object detection

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ABSTRACT

Underwater sonar image object detection is a key enabling technique for marine exploration, fisheries monitoring, and underwater robotic navigation. However, sonar imagery is often affected by severe noise interference, low contrast, blurred boundaries, and complex background clutter, leading to incomplete target representation and making scale variation difficult to characterize, which in turn limits the feature extraction and multi-scale modeling capability of conventional detectors. Meanwhile, resource-constrained underwater platforms impose strict requirements on model size and computational cost, calling for a lightweight detection solution that better balances detection accuracy and computational complexity. To this end, this paper proposes UWLight-YOLO, a lightweight object detector based on YOLOv8 for underwater sonar image analysis, aiming to reduce model complexity while maintaining competitive detection accuracy. Specifically, we first propose two lightweight C2f variants, C2f-Fast and C2f-FAMA, which replace the repeatedly stacked Bottleneck units in the original C2f with PConv-based FastBlock and EMA-recalibrated FAMABlock, respectively, thereby substantially reducing redundant computation without changing the input/output resolutions or channel configurations and improving feature representation robustness under complex sonar degradations. Subsequently, we propose the Coordinate-Enhanced Screening Feature Pyramid Network (CSFPN), which enables efficient multi-scale feature screening and fusion in the neck, thereby enhancing robustness under low signal-to-noise ratio conditions and blurred-boundary scenarios. In addition, we propose a Task Adaptive Dynamic Detection Head (TADDH) to strengthen information interaction between classification and localization, further reducing parameters and computation while improving the consistency of classification and regression predictions. Extensive experiments on the UATD dataset demonstrate that, compared with YOLOv8n, UWLight-YOLO reduces parameters by 61.3% and GFLOPs by 50.6% while maintaining comparable detection performance, validating its effectiveness and practicality as a deployment-oriented lightweight solution for underwater sonar object detection.

1. Introduction

In recent years, with the development of marine resource exploration and underwater detection technologies, research on target detection based on Forward-Looking Sonar (FLS) has gained widespread attention. Compared to optical imaging technologies, FLS offers strong penetration capability, is unaffected by water turbidity, and can detect underwater targets in low-light or completely

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dark environments [1]. As a result, FLS has broad application prospects in underwater mapping [2], shipwreck detection [3], pipeline monitoring [4], and underwater salvage [5].

However, due to the inherent characteristics of sonar imaging, underwater object detection still faces significant challenges. On one hand, sonar images are affected by marine environmental noise, seafloor reflection interference, and equipment noise, resulting in low imaging quality and poor contrast between targets and the background [6]. On the other hand, underwater targets often exhibit blurred boundaries and significant scale variations, making object detection methods developed for optical images difficult to transfer to sonar images without dedicated adaptation. Moreover, although deep learning has advanced underwater object detection, many existing methods still involve non-negligible computational and memory demands, which increases the difficulty of deployment on resource-constrained embedded underwater platforms.

The rapid advancement of deep learning has driven the evolution of underwater object detection techniques. Object detection algorithms based on Convolutional Neural Networks (CNNs), such as Faster R-CNN [7], YOLO [8], and SSD [9], have achieved remarkable success in optical image detection tasks. Among them, the YOLO (You Only Look Once) series is widely used due to its end-to-end detection framework and favorable accuracy-efficiency trade-off. Nevertheless, applying existing deep learning-based detectors to sonar images still poses several challenges. First, deployment constraints-mainstream detectors often contain a large number of parameters and require considerable computation, which may limit practical use on embedded underwater devices. Second, feature representation under sonar degradations-due to the low resolution, noise contamination, and blurred boundaries in sonar imagery, conventional architectures may struggle to extract discriminative features reliably, leading to missed detections or false alarms. Third, generalization across operating conditions-the complexity of underwater environments can cause distribution shifts across scenes and sensors, reducing model robustness when transferring to different scenarios.

To address these challenges, researchers have proposed various optimization strategies in recent years to enhance model robustness and deployment feasibility in complex underwater environments. One key direction is lightweight network design: models such as MobileNet [10], ShuffleNet [11], and EfficientNet [12] aim to reduce computational cost and model size, making them more suitable for embedded devices. In addition, feature enhancement and attention mechanisms have been introduced to improve representation quality under degraded imaging conditions. Methods such as Coordinate Attention (CA) [13], Channel Attention (SE module) [14], and Spatial Attention (CBAM) [15] enhance focus on informative regions, thereby improving detection performance. For multi-scale feature fusion, techniques such as the Feature Pyramid Network (FPN), Path Aggregation Network (PANet), and Bidirectional Feature Pyramid Network (BiFPN) integrate features across different scales to improve small-object detection and robustness to scale variations [16]. Furthermore, task-adaptive optimization has attracted increasing attention, including improvements in loss functions, data augmentation, and feature extraction modules tailored to sonar data, with the goal of mitigating noise effects and enhancing detection stability. Despite these advances, achieving a better balance between detection performance and computational efficiency for resource-constrained deployment remains challenging, motivating further investigation.

To address these limitations, this paper proposes UWLIGHT-YOLO, a lightweight object detector built upon YOLOv8, aiming to improve robustness on degraded sonar imagery while achieving a better accuracy-computational cost trade-off for resource-constrained deployment. The main contributions are summarized as follows:

(1) **Lightweight C2f variants:** C2f-Fast and C2f-FAMA. We propose two C2f-level replacements by substituting the repeatedly stacked Bottleneck units in the original C2f with PConv-based FastBlock and EMA-recalibrated FAMABlock, respectively. This design substantially reduces redundant computation without changing input/output resolutions or channel configurations, and improves feature representation robustness under degradations such as noise and blurred boundaries.

(2) **Coordinate-Enhanced Screening Feature Pyramid Network (CSFPN):** We develop an improved screening-based feature pyramid by incorporating Coordinate Attention, enabling more effective multi-scale feature fusion and enhancing robustness under low-SNR conditions.

(3) **Task Adaptive Dynamic Detection Head (TADDH):** We propose TADDH to strengthen information interaction between classification and localization, leading to more consistent predictions while reducing model complexity, thereby further supporting lightweight deployment.

The remainder of this paper is organized as follows: [Section 2](#) reviews related work, [Section 3](#) details the proposed method and its core modules, [Section 4](#) presents the experimental design and analysis of results, and [Section 5](#) concludes the paper and discusses future research directions.

2. Related work

2.1. Lightweight underwater object detection method

In marine exploration, unmanned underwater vehicle (UUV) navigation, and underwater security, underwater object detection plays a crucial role. However, due to the low contrast, noise interference, and blurred boundaries commonly found in sonar images, traditional optical image-based methods cannot be directly applied to sonar data. To balance computational complexity and detection accuracy, researchers have proposed various lightweight strategies, primarily incorporating lightweight convolutional networks such as MobileNetV2 [10], ShuffleNet [11], and EfficientNet [12] to reduce computational cost and improve compatibility with embedded devices. For example, Yang et al. proposed Fast-YOLOv5, which integrates MobileNetV2 and SENet to improve detection efficiency [17], while their FLSD-Net significantly reduces computational overhead while maintaining accuracy [18]. Wang et al. introduced YOLO-SONAR, which incorporates Coordinate Channel Attention Mechanism (CCAM) and Spatial Group Enhanced Attention Mechanism (SGEAM) to enhance target localization and small-object detection [19]. Li et al. developed DSML-YOLO, which

utilizes Depthwise Separable Convolution (DSC Block) and Lightweight Spatial Pyramid Pooling (LSPPF) to further enhance detection performance [20]. Meanwhile, Ma et al. proposed BES-YOLO, which builds on YOLOv8 and integrates Efficient Multi-scale Attention (EMA) and Bidirectional Feature Pyramid Network (BiFPN), improving multi-scale feature fusion and demonstrating superior performance in side-scan sonar image detection [21]. Although these methods achieve performance improvements, challenges remain in resource-constrained environments, where balancing detection accuracy and real-time efficiency is still difficult. To address this, this paper proposes UWLight-YOLO and constructs two lightweight C2f replacement units, namely C2f-Fast and C2f-FAMA, within the C2f modules of the YOLOv8 backbone. Specifically, C2f-Fast employs PConv-based FastBlock to achieve low-cost local structural modeling, while C2f-FAMA further introduces EMA to form FAMABlock for feature recalibration. As a result, the proposed replacements substantially reduce redundant computation without changing the input/output resolutions or channel configurations, and enhance feature representation robustness under complex sonar degradations.

2.2. Detection head optimization and efficient feature extraction

In object detection tasks, existing models typically employ a shared-feature detection head, where classification and localization tasks utilize the same feature representation. However, in complex scenarios, this shared approach can lead to information conflicts, negatively impacting detection accuracy. To address this issue, researchers have proposed various optimization strategies to improve the adaptability of detection heads to different task requirements. For example, Zhang et al. proposed an improved YOLOv5-based underwater object detection method for forward-looking sonar images, integrating transfer learning, IoU k-means anchor optimization, and the CoordConv mechanism to enhance feature extraction, thereby improving small-object detection accuracy [22]. Meanwhile, Transformer-based architectures effectively capture global features, but their high computational complexity makes real-time inference on embedded underwater devices challenging [23]. To further reduce the computational cost of the detection head, this study introduces the Task Adaptive Dynamic Detection Head (TADDH), designed to dynamically regulate information flow between classification and localization tasks. By adaptively aligning these tasks, TADDH effectively mitigates feature conflicts, thereby enhancing detection accuracy while substantially reducing model complexity and parameter count. Furthermore, this study proposes the High-level Screening Feature Pyramid Network (HSFPN), which integrates Coordinate Attention to achieve efficient multi-scale feature fusion. This architecture demonstrates superior robustness and detection performance, particularly in low-SNR environments and challenging scenarios involving blurred target boundaries.

3. Method

YOLOv8 provides a favorable balance between detection accuracy and model efficiency, and has been widely used in resource-constrained scenarios. However, directly applying it to underwater sonar image detection remains challenging. Sonar images often suffer from low signal-to-noise ratio (SNR), blurred boundaries, and complex background interference, which can weaken discriminative cues and cause the loss of key structural information. In addition, the limited compute and memory budgets on underwater platforms such as AUVs and ROVs impose strict constraints on model size and computation, making deployment of heavy networks difficult in practice. As a result, existing methods still struggle to achieve robust detection under sonar degradations while maintaining a favorable accuracy-efficiency trade-off.

To address these limitations, this paper proposes UWLight-YOLO, a lightweight and adaptive underwater object detection model built upon the YOLOv8 framework. The proposed network introduces adaptive attention and multi-scale feature fusion to improve fine-grained feature representation under noisy and low-contrast conditions. By optimizing the architecture from feature extraction, feature fusion, and detection head design, UWLight-YOLO aims to improve robustness to sonar degradations while reducing model complexity. Experimental results show that UWLight-YOLO maintains competitive detection performance with substantially reduced parameters and computation, supporting deployment on resource-constrained underwater platforms. The overall architecture is illustrated in Fig. 1.

3.1. Fast lightweight C2f module (C2f-Fast) and its attention-enhanced variant (C2f-FAMA)

For sonar-based object detection on resource-constrained underwater platforms, reducing computational complexity and parameter count is a key prerequisite for model lightweighting. To this end, the recently proposed lightweight network FasterNet introduces Partial Convolution (PConv), which applies regular convolution only to a subset of input channels for spatial feature extraction, while keeping the remaining channels unchanged and fusing information via subsequent lightweight channel mixing [24]. Inspired by this design, we lightweight the C2f module in YOLOv8: without altering the overall C2f topology, the repeatedly stacked Bottleneck units are replaced with FastBlock. The corresponding block-level definitions are shown in Fig. 2. By reducing the number of channels participating in spatial convolution, this modification effectively lowers redundant computation and memory access overhead, thereby substantially reducing the module's computational complexity and parameter count, and providing a lightweight building block for subsequent detector construction.

To quantify the efficiency gain, we estimate the convolutional cost in terms of FLOPs. Given an input feature map of spatial size $H \times W$ with C_{in} input channels and C_{out} output channels, a standard $k \times k$ convolution has an approximate complexity

$$F_{std} \approx HW k^2 C_{in} C_{out}. \quad (1)$$

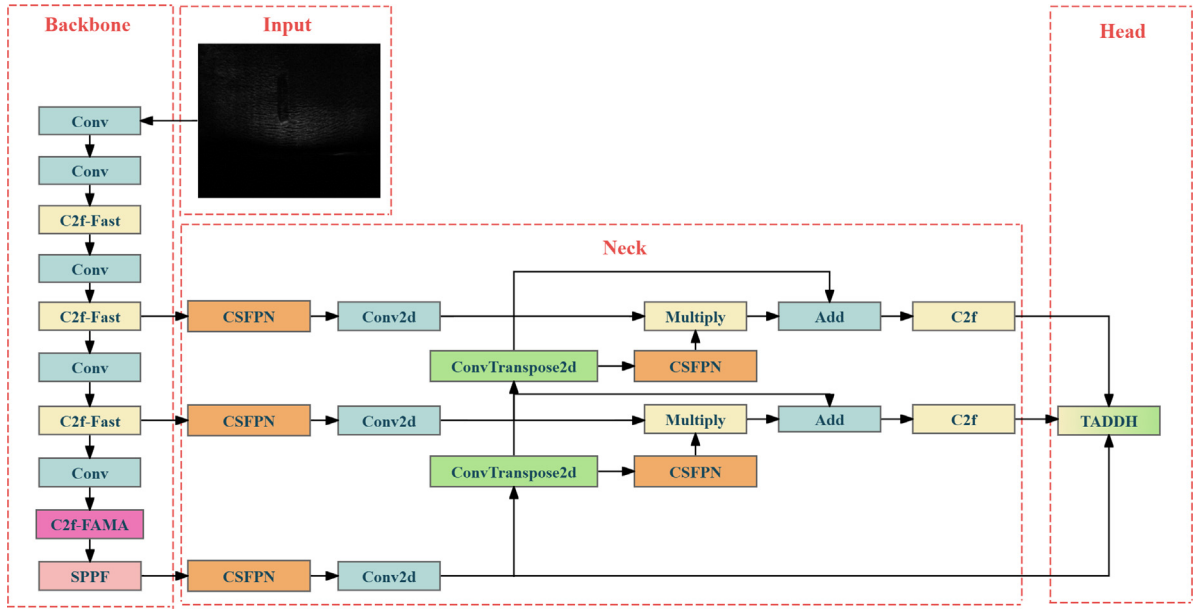


Fig. 1. The structure of UWLIGHT-YOLO.

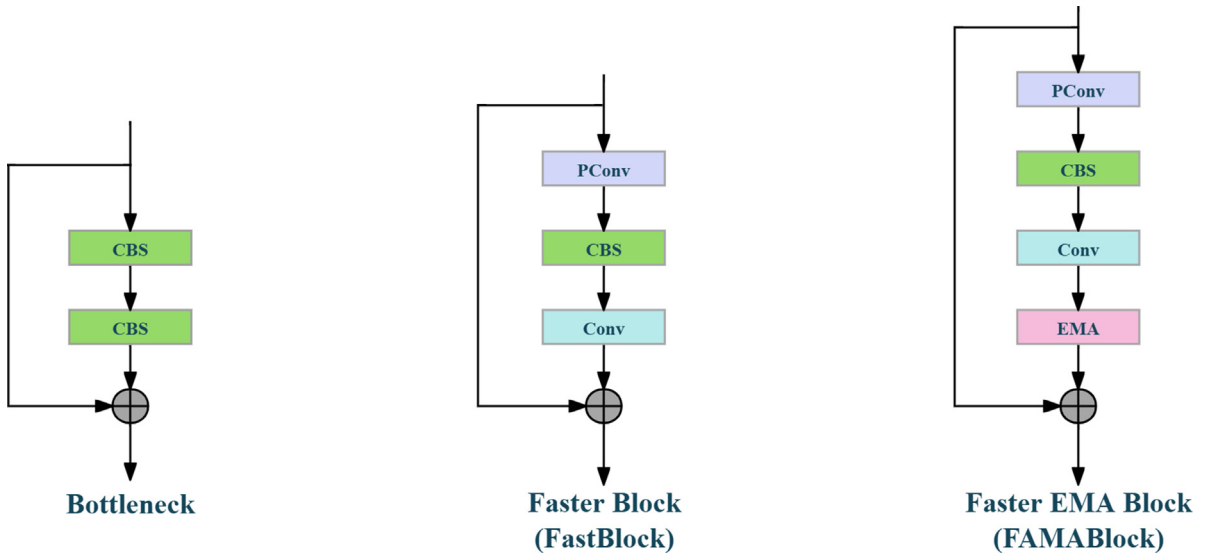


Fig. 2. Architectural definitions of Bottleneck, FastBlock, and FAMABlock.

In partial convolution (PConv), the spatial $k \times k$ operation is performed only on a subset of channels $C_p = rC_{in}$ ($0 < r < 1$), while the remaining channels bypass the spatial operator and are fused through subsequent lightweight channel mixing. Accordingly, the dominant spatial-convolution cost is reduced approximately in proportion to r , and the overall complexity can be written as

$$F_{pconv} \approx HW k^2 C_p C_{out} + F_{mix}, \quad (2)$$

where F_{mix} denotes the cost of lightweight channel mixing whose exact form depends on the implementation (e.g., pointwise/grouped operations) and is typically much smaller than the spatial $k \times k$ term in our setting. This design effectively reduces redundant computation and memory access overhead while preserving local structural modeling capacity for noisy, low-contrast sonar targets. However, under complex sonar imaging degradations, overly lightweight operators may slightly weaken feature representation. To mitigate this issue, we introduce an attention mechanism with controlled computational and parameter overhead to adaptively recalibrate sonar features, thereby enhancing discriminative responses for weak-echo targets and suppressing background clutter.

Specifically, we adopt Efficient Multi-head Attention (EMA) [25] to perform multi-head modeling and cross-dimension interaction, highlighting informative echo regions with a controllable overhead and improving representation stability. EMA aggregates multi-

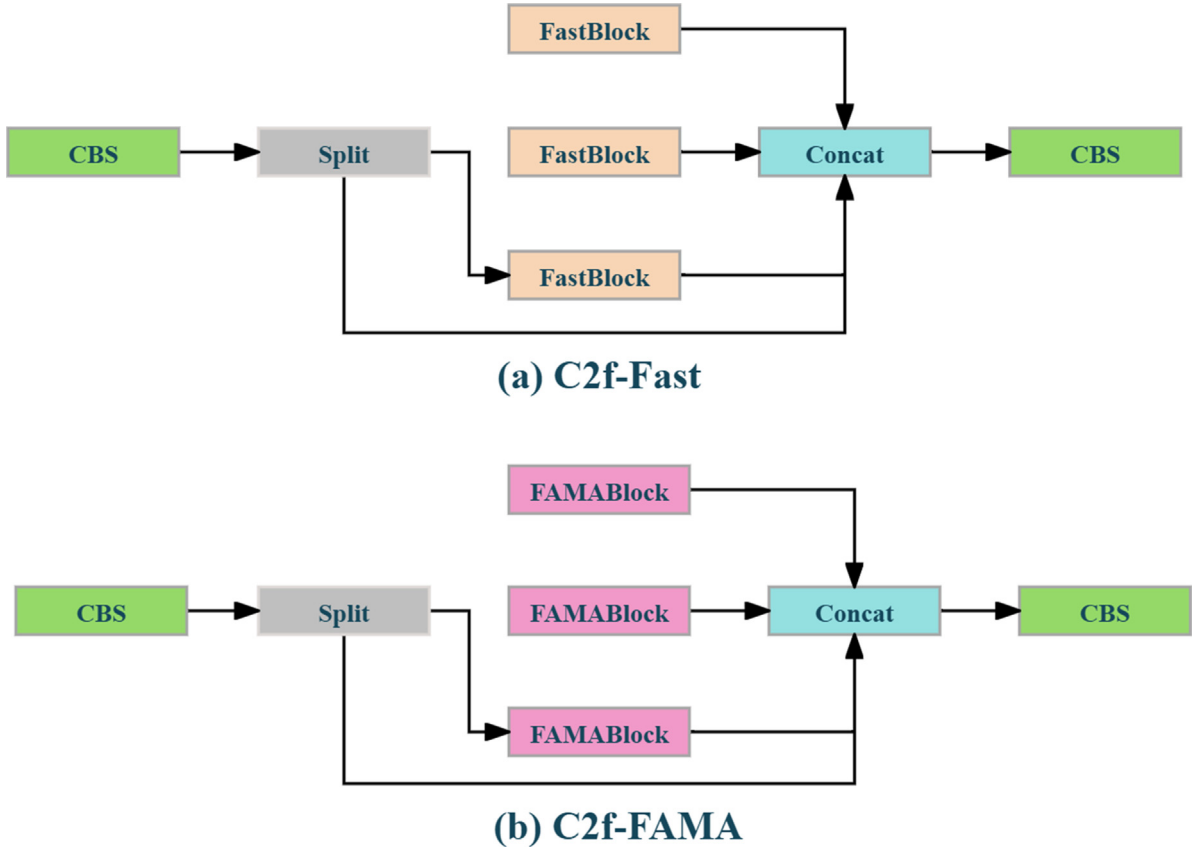


Fig. 3. The structure of C2f-Fast and C2f-FAMA.

scale spatial cues and emphasizes informative echo regions while suppressing background interference; a global descriptor commonly used for attention generation can be expressed via global average pooling (GAP) as

$$z_c = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W x_c(i, j). \quad (3)$$

Furthermore, we structurally integrate EMA with the aforementioned lightweight feature modeling unit to form a plug-and-play feature enhancement module, termed FAMABlock (see Fig. 2). By preserving local structural modeling while introducing global dependency complement from EMA, the proposed module yields more robust feature representations under severe sonar degradations and improves the stability of detection performance.

Based on the above building blocks, we instantiate them at the C2f stage, as shown in Fig. 3. C2f-Fast preserves the original C2f topology by substituting the repeated Bottleneck units in the transform branch with FastBlock (Fig. 3(a)). Further, C2f-FAMA upgrades FastBlock to the proposed FAMABlock (Fig. 3(b)), achieving lightweight local structural modeling with subsequent EMA-based feature reweighting for more robust representations. Both variants keep the same input/output resolutions and channel configurations as the original C2f, enabling backbone lightweighting without modifying the overall network-scale design.

3.2. Coordinate-enhanced screening feature pyramid network (CSFPN)

Under low-SNR sonar imaging with blurred boundaries and heavy clutter, target echoes often exhibit weak responses and sparse spatial distributions. When the neck feature fusion stage relies solely on channel-statistics-based recalibration, critical spatial positional cues can be compressed and lost, thereby weakening the characterization of target contours and structural details. We incorporate Coordinate Attention (CA) [13] into the neck design, as illustrated in Fig. 4. Unlike conventional channel attention that compresses spatial information into a single global descriptor, CA explicitly encodes coordinate-aware dependencies by aggregating features along one spatial axis while preserving positional cues along the other [13]. This enables location-sensitive feature reweighting, which is particularly beneficial for sonar imagery where targets are often submerged in cluttered backgrounds under low contrast and strong noise.

Meanwhile, CA adopts a lightweight and plug-and-play design that can be seamlessly integrated into CNN-based detectors [13]. It mainly relies on 1D pooling and a few 1×1 convolutional transformations to generate attention weights, introducing only limited

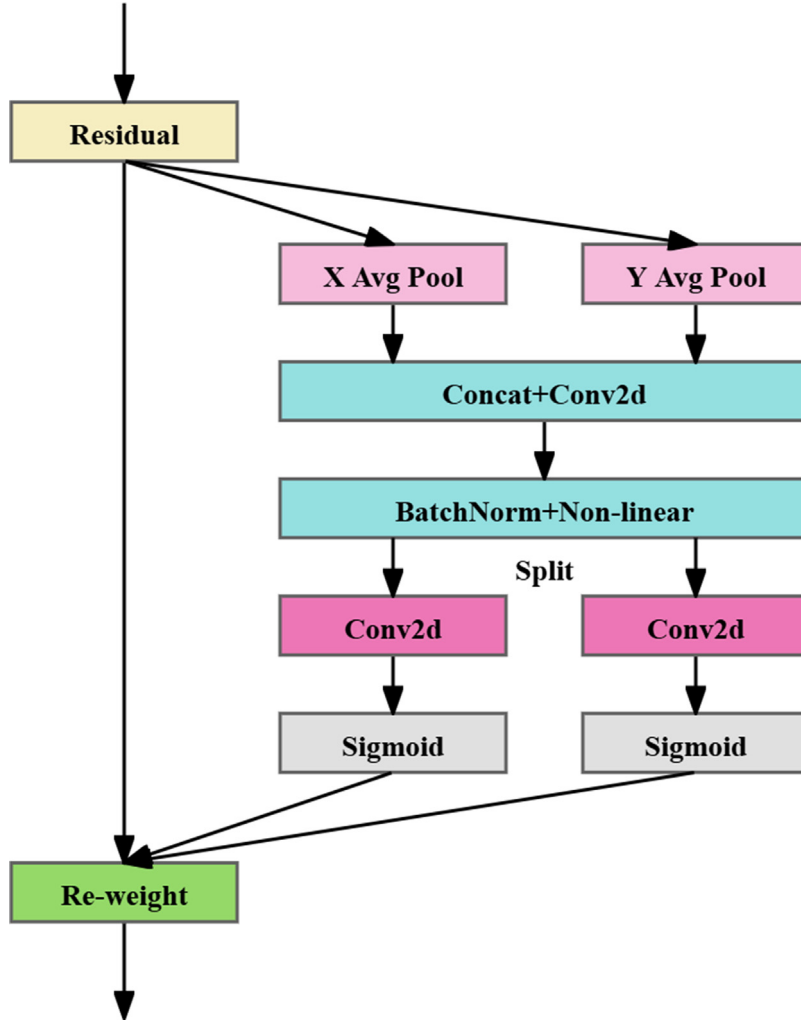


Fig. 4. Structure of the Coordinate Attention (CA) module.

parameters and computation. Compared with heavier attention mechanisms, CA is more suitable for lightweight sonar detectors deployed on resource-constrained underwater platforms, as it enhances spatial-aware representation with a controllable computational and memory overhead.

Specifically, given an input feature map $X \in \mathbb{R}^{C \times H \times W}$, where $x_c(i, j)$ denotes the feature value of channel c at spatial location (i, j) , CA performs 1D global pooling along the height and width directions to obtain two direction-aware descriptors:

$$z_c^h(i) = \frac{1}{W} \sum_{j=1}^W x_c(i, j), \quad i = 1, \dots, H, \quad (4)$$

$$z_c^w(j) = \frac{1}{H} \sum_{i=1}^H x_c(i, j), \quad j = 1, \dots, W. \quad (5)$$

The two descriptors are concatenated and transformed by a lightweight 1×1 convolution to produce an intermediate representation:

$$f = \delta(F_1([z^h, z^w])), \quad (6)$$

where $[\cdot, \cdot]$ denotes concatenation and $\delta(\cdot)$ is a nonlinear activation. Then f is split into f^h and f^w , followed by two independent transformations to generate attention weights along the two directions [13]:

$$g^h = \text{sigmoid}(F_h(f^h)), \quad g^w = \text{sigmoid}(F_w(f^w)). \quad (7)$$

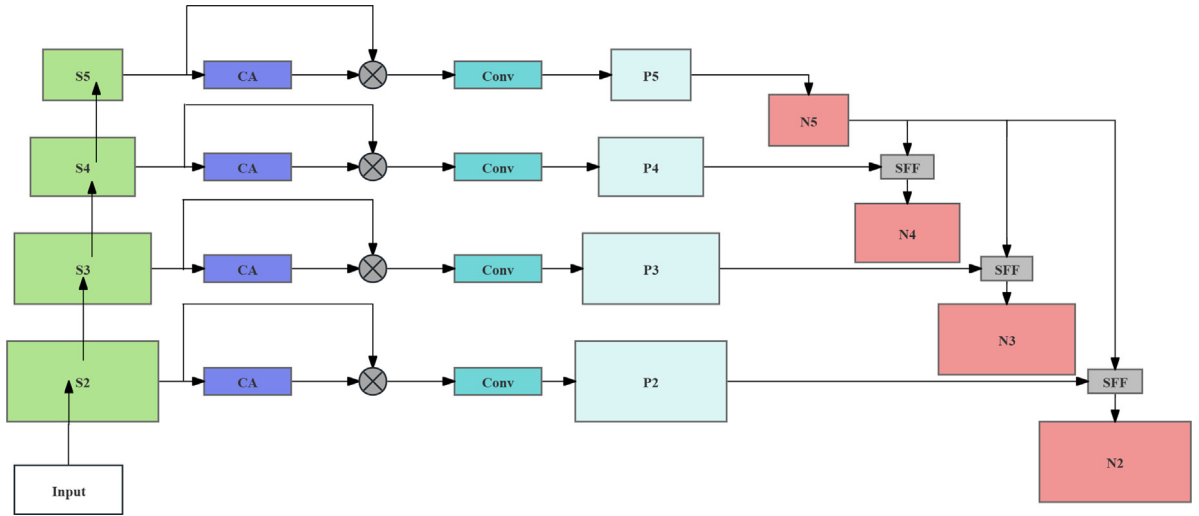


Fig. 5. The structure of CSFPN.

Finally, CA reweights the input feature map by jointly applying the horizontal and vertical attention weights:

$$y_c(i, j) = x_c(i, j) \cdot g_c^h(i) \cdot g_c^w(j). \quad (8)$$

Building upon CA, we further integrate it into the High-level Screening Feature Pyramid Network (HSFPN) [26] to construct the Coordinate-Enhanced Screening Feature Pyramid Network (CSFPN). HSFPN is a screening-based feature pyramid consisting of a feature selection module and a feature fusion module, aiming to strengthen the interaction and complementarity between high-level semantic cues and low-level structural details.

In the feature selection module, multi-scale features are first aligned via a dimension matching mechanism, and then screened by attention-guided recalibration. The original screening attention in HSFPN is mainly derived from channel statistics, which may be insufficient to explicitly encode spatial location cues under typical sonar degradations such as low contrast, blurred boundaries, and cluttered backgrounds. Therefore, we replace the original channel-attention-based screening in HSFPN with Coordinate Attention (CA) [13], introducing coordinate information into the screening process. This modification enables the screening weights to account for both channel importance and location-sensitive cues, thereby emphasizing informative echo regions while suppressing background interference, and improving the characterization of target contours and spatial distributions. Meanwhile, due to the lightweight and plug-and-play nature of CA, the proposed enhancement strengthens feature selection and fusion with controllable parameter and computational overhead, which aligns well with the lightweight design goals and deployment constraints of underwater platforms.

Based on the above design, we embed CA into HSFPN to construct CSFPN for multi-scale screening and fusion in the neck. The overall architecture of CSFPN is illustrated in Fig. 5. CSFPN aligns multi-level features and performs screening-based recalibration, and then aggregates cross-scale information along a top-down fusion pathway, ultimately producing pyramid features for the subsequent detection head.

3.3. Task adaptive dynamic detection head (TADDH)

In sonar object detection, classification relies more on semantic discrimination, whereas localization depends more on geometric boundaries and shape details; when targets exhibit irregular contours and large scale variations, conventional decoupled heads often fail to preserve the complementarity between the two branches, leading to degraded prediction consistency. We propose the Task Adaptive Dynamic Detection Head (TADDH), as illustrated in Fig. 6. Targeting typical challenges in sonar imaging, including low signal-to-noise ratio, irregular target shapes, and large scale variation, TADDH integrates shared feature interaction, task-decomposition-based feature allocation, and dynamic deformable geometric modeling within the detection head. In this way, the classification and localization branches are deeply coordinated along a share–allocate–re-interact pathway, enabling a favorable trade-off between computational efficiency and practically usable detection accuracy.

Specifically, TADDH takes the multi-level pyramid features $\{P_3, P_4, P_5\}$ from the neck as inputs, corresponding to targets at different scales. It first employs several shared 3×3 convolutional layers to extract primary discriminative features and facilitate information interaction between the two tasks. Group Normalization is incorporated to stabilize optimization and improve convergence consistency between classification and localization. On top of the shared representation, TADDH adopts a dynamic Task Decomposition (TD) module for task-adaptive feature allocation [27]. The key idea is to generate lightweight attention-like allocation weights and split the shared features into task-preferred components: category-discriminative features for classification and boundary-/geometry-related features for regression, thereby suppressing feature conflict while retaining cross-task interaction.

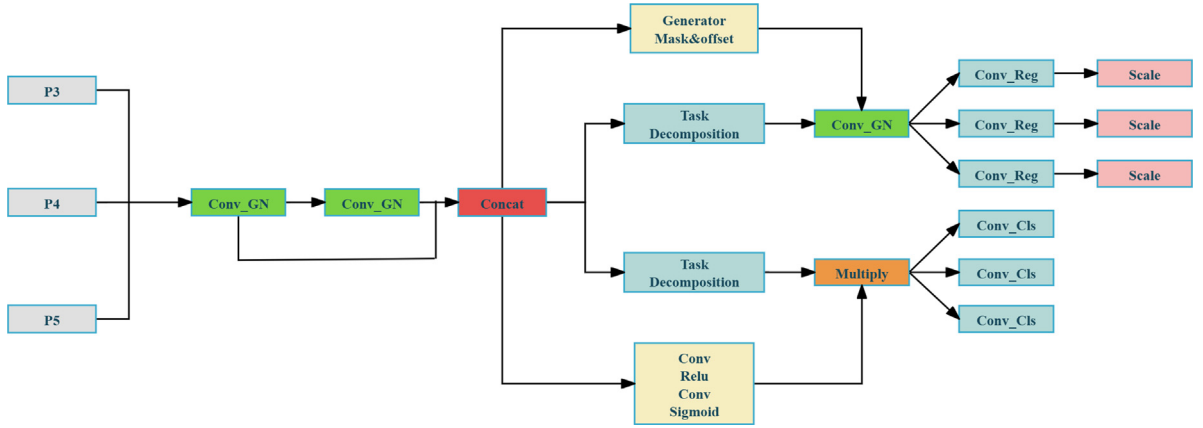


Fig. 6. The structure of TADDH.

Implementation details. For each pyramid level $P_i \in \{P_3, P_4, P_5\}$, the input feature is denoted as $x_i \in \mathbb{R}^{h_{dc} \times H_i \times W_i}$, where $h_{dc} = 256$ in our implementation. The shared interaction branch consists of two 3×3 Conv_GN layers with channel width $h_{dc} \rightarrow h_{dc}/2 \rightarrow h_{dc}/2$ (i.e., $256 \rightarrow 128 \rightarrow 128$), and their outputs are concatenated along channels, producing the shared interaction feature $F_i \in \mathbb{R}^{C \times H_i \times W_i}$ with $C = 256$. For the TD module, we adopt a lightweight two-layer bottleneck with an intermediate width of 16 to generate the allocation weights for both classification and regression branches.

Let the shared feature map after concatenation be $F_i \in \mathbb{R}^{C \times H_i \times W_i}$. TD first aggregates global context by compressing the spatial dimensions using adaptive global average pooling:

$$F'_i = \text{AdaptiveAvgPool2d}(F_i; 1 \times 1). \quad (9)$$

Then, TD generates a task allocation weight vector w_i via stacked convolutions and a sigmoid activation:

$$w_{i,1} = \text{Conv}(F'_i; W_1), \quad w_i = \sigma(\text{Conv}(w_{i,1}; W_2)), \quad (10)$$

where $\sigma(\cdot)$ denotes the sigmoid function and W_1, W_2 are convolution parameters. Finally, TD performs complementary reweighting and projects the features into task-specific representations for classification and regression:

$$F_{cls,i} = \text{Conv}(F'_i \odot w_i; W_{cls}), \quad F_{reg,i} = \text{Conv}(F'_i \odot (1 - w_i); W_{reg}), \quad (11)$$

where $\odot$ denotes element-wise multiplication and W_{cls}, W_{reg} are the projection parameters for the classification and regression branches, respectively. This decomposition encourages the classification branch to focus on category-discriminative cues, while allowing the regression branch to emphasize boundary- and shape-related information, thus structurally alleviating the optimization instability induced by forcing a single feature to serve both tasks.

Furthermore, to handle irregular contours and viewpoint-dependent geometric deformation and local blurring in sonar imagery, TADDH introduces dynamic deformable convolution (DyDCNv2) in the localization branch [28]. Through interactive offset learning, semantic context from the classification branch is injected into the regression branch to predict more reliable boundaries for irregular targets. Concretely, the interaction features are used to estimate sampling offsets Δp_k and modulation masks m_k , which adaptively adjust the regular sampling locations p_k and dynamically reshape the receptive field to better fit target geometry. In a standard DCNv2 form, this process can be expressed as

$$y(p_0) = \sum_{k=1}^K w_k \cdot x(p_0 + p_k + \Delta p_k) \cdot m_k, \quad (12)$$

where $x(\cdot)$ and $y(\cdot)$ denote the input and output features, and w_k is the convolution kernel weight. In our implementation, DyDCNv2 uses a 3×3 kernel; offsets and masks are predicted by a 3×3 convolution producing 27 channels, which are split into an 18-channel offset ($2k^2$) and a 9-channel mask (k^2) with sigmoid normalization. By learning adaptive sampling and modulation, this mechanism strengthens the modeling of edges, contours, and local details, improving localization adaptability across different imaging angles and shape variations.

Meanwhile, to enhance the sensitivity of the classification branch to multi-scale targets, especially small targets and distant weak-echo targets, TADDH performs dynamic feature selection via a lightweight spatial gating branch. Specifically, a probability map $p_i \in \mathbb{R}^{1 \times H_i \times W_i}$ is predicted from F_i using a 1×1 convolution ($256 \rightarrow 64$) followed by a 3×3 convolution ($64 \rightarrow 1$) and a sigmoid activation, and the classification feature is modulated as $F_{cls,i} \odot p_i$ to improve robustness under scale changes. In the regression branch, a Scale Layer is applied for scale calibration: we adopt a per-level learnable scalar $\text{Scale}(\cdot)$ (initialized to 1.0) to rescale the regression logits, improving the consistency of boundary regression across targets of different sizes.

In summary, by synergistically combining shared convolutional interaction, TD-based dynamic feature allocation, DyDCNv2-driven geometric adaptation, dynamic feature selection, and Scale-layer calibration, TADDH reduces redundant computation at the

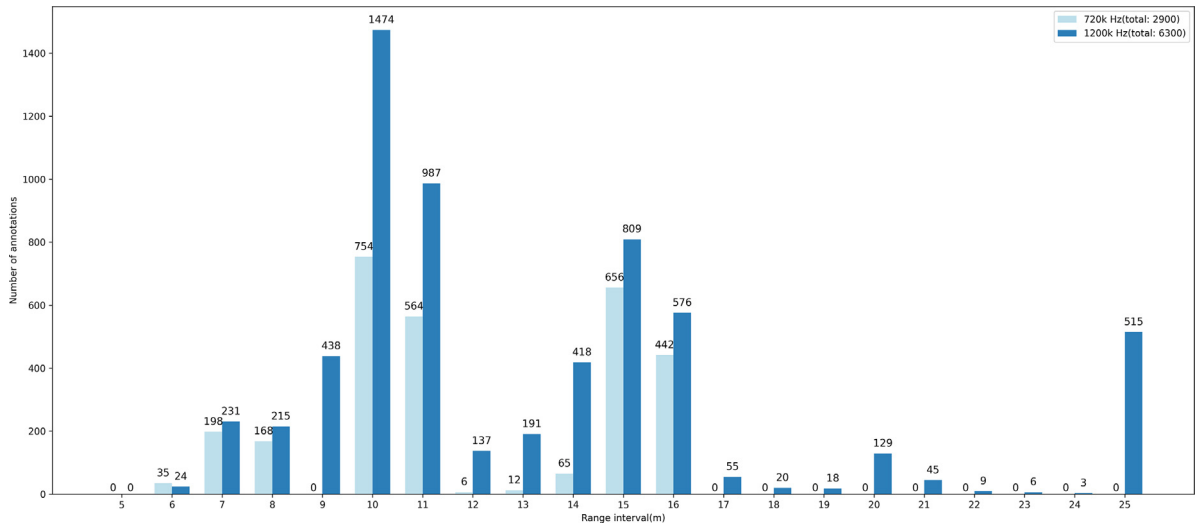


Fig. 7. Image count statistics over range intervals under two operating frequencies.

head level and improves task consistency. This makes it more suitable for deployment on resource-constrained underwater platforms, while exhibiting stronger robustness to small, weak-echo, and irregular sonar targets in complex scenes.

4. Experiments

4.1. Datasets and experimental setup

This study evaluates the proposed model on the Underwater Acoustic Target Detection (UATD) dataset [29], a publicly available benchmark designed for multi-beam forward-looking sonar (MFLS) target detection. UATD contains 9200 BMP-format sonar images with corresponding XML annotations for 10 object categories. The data were acquired at two operating frequencies, including 2900 images collected at 720 kHz and 6300 images collected at 1200 kHz. During acquisition, the sonar working range spans 5–25 m; therefore, UATD provides the image-count statistics binned into 1 m range intervals. Fig. 7 shows the numbers of images collected at 720 kHz and 1200 kHz, respectively.

UATD provides an official split of 7,600/800/800 images for training/validation/test, which we adopt without any re-partitioning to ensure fair and reproducible evaluation. Notably, the validation and test sets exhibit noticeable differences from the training set in terms of scene distribution and target imaging characteristics, which facilitates a more comprehensive assessment of detection accuracy and generalization and helps mitigate overfitting to the training scenarios. Meanwhile, Fig. 8 illustrates the class-wise distribution of annotated targets in UATD.

The proposed algorithm was trained and evaluated on an NVIDIA Quadro RTX 8000 GPU with 48 GB VRAM. The software environment includes Python 3.8, PyTorch 1.13.1, and CUDA 11.7. We adopt SGD as the optimizer for network training.

4.2. Evaluation metrics

To provide a comprehensive evaluation of the proposed UWLight-YOLO, we adopt a set of standard detection accuracy metrics and lightweightness-related metrics, including Mean Average Precision (mAP), Precision (P), Recall (R), the number of parameters, computational complexity (GFLOPs), and inference throughput (FPS). mAP summarizes detection accuracy across categories, while P and R characterize the tendencies of false positives and false negatives, respectively. The number of parameters reflects the model size in terms of trainable weights, and GFLOPs measures computational cost, where smaller values generally indicate lower resource consumption and better suitability for resource-constrained platforms. FPS denotes the end-to-end inference throughput under a unified evaluation setting and is used to assess runtime behavior and real-time feasibility. Together, these metrics quantitatively characterize the detection performance of UWLight-YOLO and its trade-off among accuracy, parameter count, and computational complexity.

4.3. Ablation study

To evaluate the effectiveness of the proposed modules, we conduct ablation studies on the UATD dataset, using the original YOLOv8n as the baseline. We first perform backbone-level ablations to investigate the effects of lightweight operator substitution and attention-enhanced feature modeling for sonar imagery. For clarity, we denote the four C2f stages in the YOLOv8n backbone

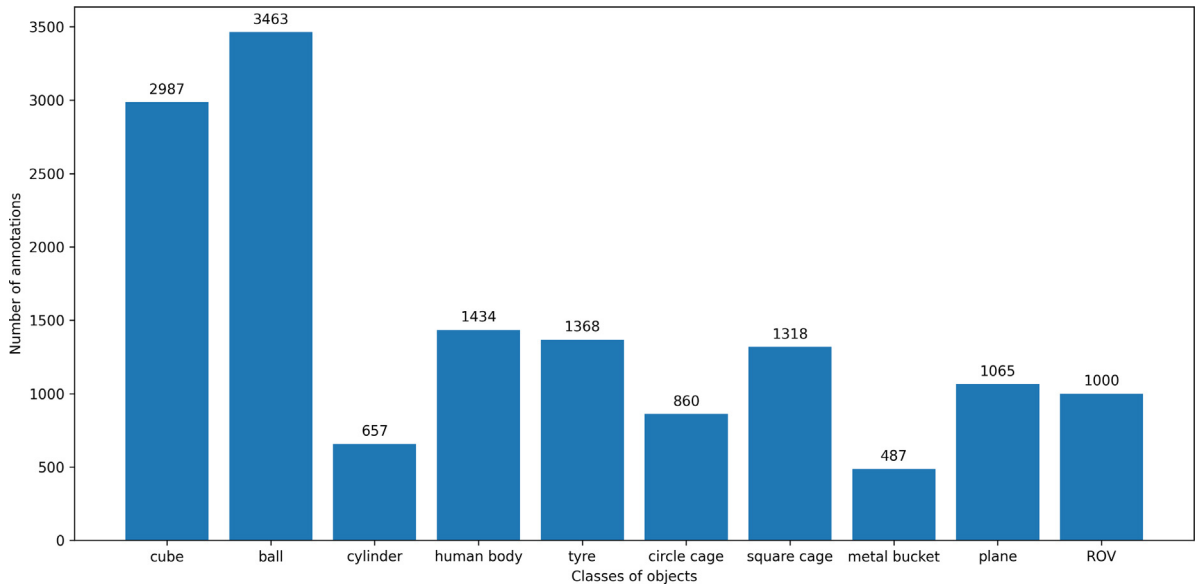


Fig. 8. Class-wise distribution of annotated targets in the UATD dataset.

Table 1
Backbone-level ablations on UATD.

Model	mAP@0.5 (%)	mAP@0.5:0.95 (%)	FPS	Params (M)	GFLOPs
Baseline	85.8	38.4	573.3	3.1	8.1
B1 (Fast-All)	85.2	38.2	611.8	2.3	7.1
B2 (Fast-Front3)	85.0	37.1	605.3	2.7	7.2
B3 (FAMA-DeepOnly)	85.1	38.3	593.4	3.1	7.8
B4 (Ours)	86.2	38.7	621.1	2.4	7.2

from shallow to deep as $\{S_1, S_2, S_3, S_4\}$, and use a configuration vector $\mathbf{b} = [b_1, b_2, b_3, b_4]$ to specify the block type used at each stage, where $b_i \in \{\text{C2f}, \text{C2f-Fast}, \text{C2f-FAMA}\}$.

Based on this notation, we construct the following backbone variants: B1 (Fast-All) uses $\mathbf{b} = [\text{C2f-Fast}, \text{C2f-Fast}, \text{C2f-Fast}, \text{C2f-Fast}]$; B2 (Fast-Front3) uses $\mathbf{b} = [\text{C2f-Fast}, \text{C2f-Fast}, \text{C2f-Fast}, \text{C2f}]$; B3 (FAMA-DeepOnly) uses $\mathbf{b} = [\text{C2f}, \text{C2f}, \text{C2f}, \text{C2f-FAMA}]$; and B4 (Ours) uses $\mathbf{b} = [\text{C2f-Fast}, \text{C2f-Fast}, \text{C2f-Fast}, \text{C2f-FAMA}]$. Here, C2f-Fast improves efficiency by reducing redundant spatial computation, while C2f-FAMA enhances representation robustness at the deepest stage. All variants are trained with identical protocols and evaluated by mAP, Params, GFLOPs, and FPS. The quantitative results are summarized in Table 1.

As shown in Table 1, the backbone variants exhibit a clear accuracy–efficiency trade-off. Applying C2f-Fast to all stages in B1 substantially reduces model complexity, with the parameter count decreasing from 3.1M to 2.3M and GFLOPs decreasing from 8.1 to 7.1, while the accuracy only slightly drops, with mAP@0.5 decreasing from 85.8 to 85.2 and mAP@0.5:0.95 decreasing from 38.4 to 38.2. This confirms the effectiveness of lightweight operator substitution. Replacing only the first three stages with C2f-Fast in B2 reduces complexity to 2.7M parameters and 7.2 GFLOPs, but it causes a notable drop in mAP@0.5:0.95 from 38.4 to 37.1, indicating that localization robustness is particularly sensitive to aggressive lightweighting in shallow-to-mid layers under sonar degradations. In contrast, introducing C2f-FAMA only at the deepest stage in B3 largely preserves localization quality with limited additional module overhead: mAP@0.5:0.95 changes from 38.4 to 38.3, while GFLOPs still decrease from 8.1 to 7.8, supporting the role of deep-stage enhancement for robust high-level representations. Combining both designs in B4 achieves the best overall trade-off, where mAP@0.5 increases to 86.2 and mAP@0.5:0.95 increases to 38.7, while the parameter count decreases to 2.4M and GFLOPs decrease to 7.2. The slightly higher complexity of B4 compared with B1 is mainly due to the lightweight recalibration introduced at the deepest stage, which improves high-level feature robustness and yields consistent gains in both mAP metrics. Moreover, B4 attains the highest FPS of 621.1 under the same benchmarking protocol, indicating favorable runtime efficiency. Based on these results, we select B4 as the default backbone configuration for the following experiments.

After selecting B4 as a more effective backbone configuration, we further conduct component-level ablations to quantify the independent contributions of each improvement module and analyze their combined effects. Specifically, we use YOLOv8n as the baseline and consider three key components: the backbone redesign B4, the CSFPN neck, and the TADDH detection head. In total, we evaluate eight settings: Baseline, Baseline + B4, Baseline + CSFPN, Baseline + TADDH, Baseline + B4 + CSFPN, Baseline + B4 + TADDH, Baseline + CSFPN + TADDH, and Ours (Baseline + B4 + CSFPN + TADDH). Note that “+” denotes component substitution: the proposed module replaces its corresponding counterpart in the baseline, rather than simply adding extra branches or layers. All models

Table 2
Ablation study on different components.

Model	mAP@0.5 (%)	mAP@0.5:0.95 (%)	FPS	Params (M)	GFLOPs
Baseline	85.8	38.4	573.3	3.1	8.1
Baseline + B4	86.2	38.7	621.1	2.4	7.2
Baseline + CSFPN	86.7	39.3	632.7	1.9	6.9
Baseline + TADDH	85.1	38.6	530.2	2.2	7.2
Baseline + B4 + CSFPN	86.4	39.2	626.9	1.6	6.3
Baseline + B4 + TADDH	85.3	38.3	502.1	1.9	7.6
Baseline + CSFPN + TADDH	86.0	39.1	608.9	1.5	5.1
Ours	85.8	39.8	537.0	1.2	4.0

Table 3
Comparison of object detectors on the UATD dataset.

Model	P(%)	R(%)	mAP@0.5(%)	mAP@0.5:0.95(%)	FPS	Params(M)	GFLOPs
SSD	81.3	79.7	80.2	35.8	520.6	13.6	15.6
Faster R-CNN	84.5	82.3	83.5	36.7	440.1	41.8	182.3
YOLOv8n	86.3	82.2	85.8	38.4	573.3	3.1	8.1
YOLOv10n	85.6	82.0	83.8	37.2	648.1	2.2	6.5
YOLOv11n	87.1	83.1	84.1	37.5	581.2	2.5	6.3
DETR	85.9	82.6	85.3	38.1	518.8	41.6	81.6
Deformable DETR	86.2	83.3	85.6	38.7	324.5	40.1	169.0
Ours	87.4	83.5	85.8	39.8	537.0	1.2	4.0

are trained and tested under identical protocols, and we report mAP, the number of parameters, GFLOPs, and FPS as evaluation metrics. The results are summarized in Table 2.

As shown in Table 2, the component-level ablation results demonstrate that the proposed modules are mainly tailored for lightweight modeling, substantially reducing model size and computational cost while keeping the overall detection accuracy stable. After introducing the B4 backbone, mAP@0.5 increases from 85.8% to 86.2% and mAP@0.5:0.95 increases from 38.4% to 38.7%. Meanwhile, the number of parameters decreases from 3.1M to 2.4M and GFLOPs drop from 8.1 to 7.2, indicating that the backbone redesign reduces complexity without compromising feature representation.

After replacing the neck with CSFPN, mAP@0.5 improves to 86.7% and mAP@0.5:0.95 improves to 39.3%. The model requires 6.9 GFLOPs, while the parameter count is further reduced to 1.9M, indicating that the proposed multi-scale screening-and-fusion neck is a parameter-efficient redesign that brings consistent accuracy gains without inflating model size. When combining B4 with CSFPN, the model achieves 86.4% mAP@0.5 and 39.2% mAP@0.5:0.95, while further reducing the parameters to 1.6M and GFLOPs to 6.3. This verifies the complementary effects of backbone lightweighting and neck-level screening-and-fusion in improving the accuracy–complexity trade-off.

In contrast, TADDH mainly serves to reduce model size and computational cost through a more compact, task-aligned detection head. With TADDH alone, the number of parameters decreases from 3.1M to 2.2M and GFLOPs drop from 8.1 to 7.2, while the detection accuracy remains comparable to the baseline. When combined with other components, the measured complexity may show small fluctuations due to operator-level and implementation-dependent factors, but the overall budget continues to decrease as the full design is enabled. More importantly, when all modules are activated, the model maintains mAP@0.5 at 85.8% while improving mAP@0.5:0.95 to 39.8%. This indicates that the joint design primarily enhances localization quality under stricter IoU thresholds, rather than merely boosting coarse detection at IoU=0.5. Such improvement can be attributed to the task-aligned design and screening-based feature refinement: the former strengthens the consistency between classification and regression, while the latter highlights informative features and suppresses redundant interference, leading to tighter box regression and more conservative yet better-calibrated confidence estimation. Consequently, the gain is more pronounced on mAP@0.5:0.95, whereas mAP@0.5 saturates and remains stable. Meanwhile, the parameters are further reduced to 1.2M and GFLOPs to 4.0, achieving a significant complexity reduction with negligible impact on mAP@0.5. Overall, compared with the baseline, UWLight-YOLO reduces the parameter count by 61.3% and GFLOPs by 50.6% while maintaining stable accuracy, making it more suitable for deployment on resource-constrained platforms.

4.4. Comparative experiments

After the ablation studies validate the effectiveness of the proposed modules, we further conduct comparative experiments on UATD to benchmark UWLight-YOLO against representative detectors from different paradigms. The compared methods include one-stage CNN-based detectors SSD [9], YOLOv8n, YOLOv10n [30], and YOLOv11n [31], the two-stage detector Faster R-CNN [7], as well as Transformer-based detectors DETR [32] and Deformable DETR [33]. For fair comparison, all methods are evaluated on the test set using the same data split and a unified benchmarking protocol, and the results are summarized in Table 3.

As shown in Table 3, UWLight-YOLO demonstrates strong detection performance and clear lightweight advantages on the UATD test set. Our method achieves 87.4% precision and 83.5% recall, reaches 85.8% mAP@0.5, and attains 39.8% mAP@0.5:0.95. Notably,

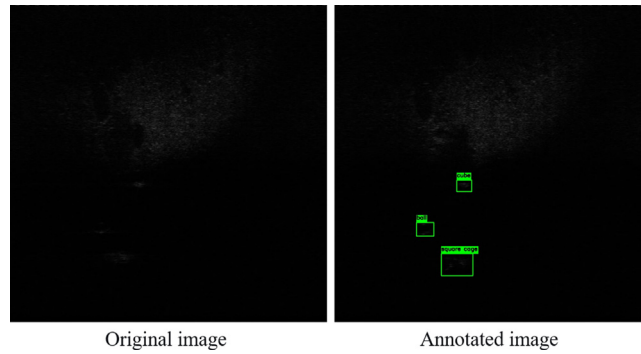


Fig. 9. Representative original sonar images and their corresponding ground-truth annotations.

UWLight-YOLO obtains the highest $\text{mAP}@0.5:0.95$ among all compared methods, indicating superior localization quality and overall detection capability under stricter IoU thresholds. Compared with the baseline YOLOv8n, UWLight-YOLO maintains a comparable $\text{mAP}@0.5$ while improving $\text{mAP}@0.5:0.95$ from 38.4% to 39.8%. Meanwhile, it compresses the parameter count from 3.1M to 1.2M and reduces GFLOPs from 8.1 to 4.0, verifying that the proposed lightweight design effectively controls model size and computational cost while preserving or even improving detection performance.

Further comparisons across different paradigms show that two-stage and Transformer-based detectors typically incur much larger parameter sizes and higher computational costs. For example, Faster R-CNN uses 41.8M parameters with 182.3 GFLOPs, DETR uses 41.6M parameters with 81.6 GFLOPs, and Deformable DETR uses 40.1M parameters with 169.0 GFLOPs. Despite their substantially higher complexity than our method, their $\text{mAP}@0.5:0.95$ are 36.7%, 38.1%, and 38.7%, respectively, all lower than UWLight-YOLO at 39.8%. Meanwhile, among lightweight one-stage baselines, YOLOv10n and YOLOv11n have 2.2M and 2.5M parameters with 6.5 and 6.3 GFLOPs, respectively, whereas UWLight-YOLO further compresses the model to 1.2M parameters and reduces computation to 4.0 GFLOPs, while achieving a higher $\text{mAP}@0.5:0.95$ and maintaining competitive $\text{mAP}@0.5$.

In addition, regarding inference speed (FPS), we report FPS for all methods under the same benchmarking protocol. Although UWLight-YOLO achieves a slightly lower FPS than the baseline YOLOv8n, it still maintains a high inference throughput that is sufficient for practical deployment. It should be emphasized that end-to-end runtime is not solely determined by GFLOPs; it is also influenced by operator implementations, memory access and data movement overhead, as well as hardware-dependent optimization strategies, especially when dynamic or non-standard operators (e.g., deformable convolution) are involved. Therefore, GFLOPs can be used as a hardware-agnostic proxy for computational complexity, while the actual end-to-end speed should be evaluated with runtime measurements such as FPS. Overall, UWLight-YOLO substantially reduces model size and computational cost while retaining strong detection performance, achieving a better accuracy–efficiency trade-off and facilitating deployment on resource-constrained underwater platforms.

After the quantitative comparisons, we further conduct qualitative analysis to examine the detection differences among various methods under challenging sonar imaging conditions, and present representative visualizations. As shown in Fig. 9, we provide raw sonar images together with the corresponding ground-truth annotations as references for visual inspection. Based on these examples, Fig. 10 summarizes the qualitative detection results of multiple competing methods and the proposed UWLight-YOLO, enabling a direct comparison of their outputs in typical cases such as cluttered backgrounds, pseudo-echo interference, and weak-echo targets.

As shown in Fig. 9, this sonar image contains three targets: cube, ball, and square cage. Among them, the square cage exhibits a weaker echo response with incomplete boundary structures and is heavily mixed with background clutter, forming a challenging detection case characterized by weak echoes, low contrast, and blurred structural details. Fig. 10 presents the qualitative comparisons across different methods. It can be observed that SSD detects only the cube and ball and misses the square cage. Faster R-CNN is able to detect the square cage, but the confidence is relatively low and the predicted box is conservative. For the one-stage YOLO family, YOLOv8n, YOLOv10n, and YOLOv11n produce stable detections on cube and ball, yet exhibit evident category confusion on the third target by misclassifying the square cage as circle cage with low confidence, indicating that category discrimination remains unstable under weak echoes and degraded structural details. For Transformer-based detectors, DETR also predicts the target as circle cage and thus suffers from misclassification. Deformable DETR recognizes it as square cage, but the confidence is still limited. In contrast, UWLight-YOLO detects all three targets and correctly identifies the third one as square cage, producing tighter localization and a more stable confidence estimate.

Overall, the qualitative visualizations further demonstrate that UWLight-YOLO provides more reliable target detection and category discrimination under weak-echo and cluttered-background conditions. It effectively alleviates missed detections and category confusion, and outputs more stable localization and confidence scores, which offers intuitive evidence supporting the robustness and practicality of the proposed method in complex sonar scenes.

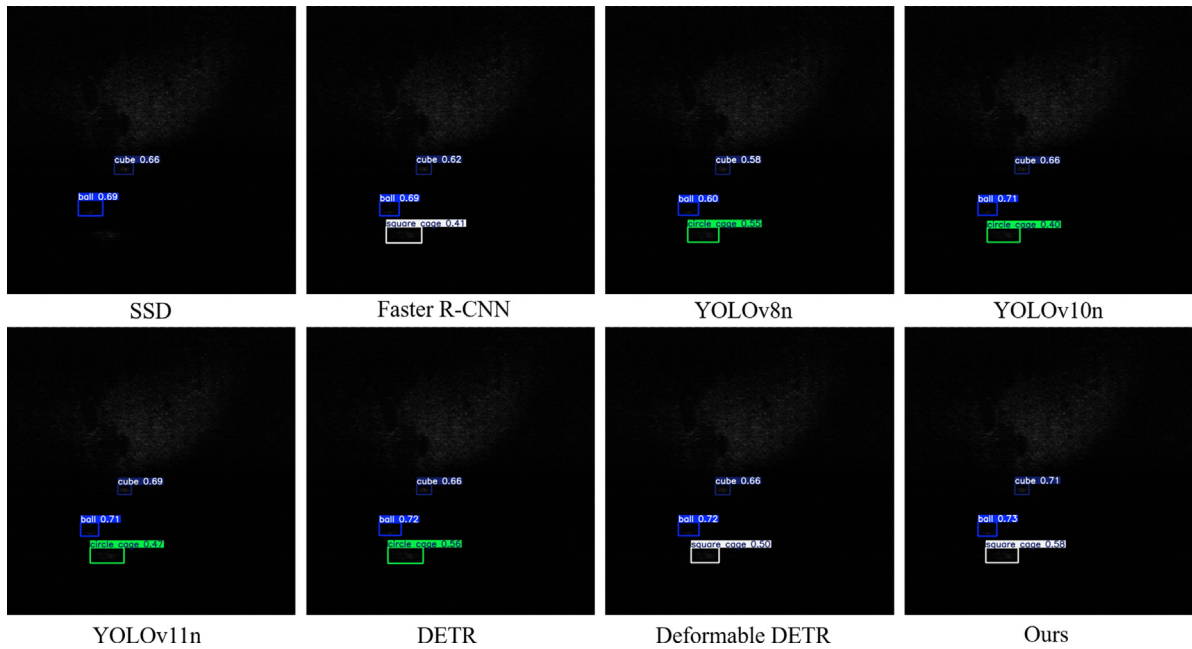


Fig. 10. Qualitative comparison of detection results across different methods.

5. Conclusion

This paper proposes a lightweight detection algorithm, UWLIGHT-YOLO, for underwater forward-looking sonar target detection, achieving a favorable balance between detection accuracy and model lightweighting. Key components of UWLIGHT-YOLO include C2f-Fast and C2f-FAMA for reducing computational overhead while maintaining robust feature representations, CSFPN for enhancing valid echo cues across scales and suppressing background clutter, and TADDH for improving classification–localization consistency to increase prediction reliability. Experiments on the public UATD dataset show that UWLIGHT-YOLO maintains mAP@0.5 of 85.8% while improving mAP@0.5:0.95 to 39.8%, and compresses the model to 1.2M parameters and 4.0 GFLOPs, substantially reducing parameter size and computational complexity compared with the YOLOv8n baseline. Furthermore, comparisons with representative one-stage, two-stage, and Transformer-based detection algorithms demonstrate that UWLIGHT-YOLO preserves competitive detection accuracy under significantly reduced model parameters and computation, and ablation studies verify the effectiveness and complementarity of the proposed components.

For future work, we will extend UWLIGHT-YOLO toward broader applicability and stronger deployment readiness. On the one hand, we will improve its generalization across diverse sonar imaging modalities (including forward-looking sonar, side-scan sonar, and multi-beam sonar) as well as more challenging operating conditions. On the other hand, we will explore self-supervised and cross-modal learning to reduce reliance on labeled data and enhance transferability. Meanwhile, we will pursue deployment-oriented optimizations on embedded hardware to further reduce end-to-end latency and improve real-time feasibility for autonomous underwater exploration systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Chenghao Ji: Methodology, Software, Investigation, Formal analysis, Writing – original draft, Visualization. **Yujie Li:** Conceptualization, Supervision, Funding acquisition, Project administration, Writing – review & editing.

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